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Wood post bracket

Abstract

A wood post bracket used for securing a post mount to a rim joist and a mounting blocks of a substructure of a deck's wood framing system includes a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls, a planar base forming a first wall of the four walls and a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate and the planar base and the mounting plate are oriented parallel to each other.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation in part of U.S. patent application Ser. No. 17/027,218, filed Sep. 21, 2020, which is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to railing systems and relates more specifically to support hardware for use in railing systems.

BACKGROUND OF THE DISCLOSURE

(2) One common style of decking system comprises vertically oriented posts that are supported by a wood substructure or framing system that includes joists. The posts, in turn, support a railing which may be oriented in a manner parallel to the joists or in a manner that is angled relative to the joists (e.g., as may be the case on a railing that is adjacent to stairs). The posts are typically coupled to the joists by fasteners (e.g., lag bolts or lag screws) which are lagged through the posts and joists (e.g., drilled directly through, without any pre-drilled holes being formed in either the posts or the joists).

SUMMARY OF THE INVENTION

(3) In one example, a wood post bracket used for securing a post mount to a rim joist and a

mounting blocks of a substructure of a deck's wood framing system includes a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls, a planar base forming a first wall of the four walls and a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate and the planar base and the mounting plate are oriented parallel to each other.

(4) In another example, a wood post bracket used for securing a post mount to a rim joist and a mounting blocks of a substructure of a deck's wood framing system includes a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls; a planar base forming a first wall of the four walls, wherein the planar base comprises, a first flange that extends away from the planar base in an orientation parallel to the hollow sleeve, such that the planar base has an L-shaped cross section, and a first plurality of apertures formed in the first flange; and a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate, wherein the mounting plate comprises a second flange that extends away from the mounting plate in a direction toward the first flange and in an orientation parallel to the hollow sleeve, such that the mounting plate has an L-shaped cross section, a second plurality of apertures formed in the second flange, a third plurality of apertures formed in a surface of the mounting plate that is oriented parallel to the planar base, and a lip positioned on an opposite end of the mounting plate from the second flange and extending past the hollow sleeve, such that the mounting plate is longer than the planar base.

(5) In another example, a deck substructure includes a rim joist coupled between a first deck joist and a second deck joist; a wood post bracket coupled to the rim joist, the wood post bracket comprising a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls, a planar base forming a first wall of the four walls, a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate and the planar base and the mounting plate are oriented parallel to each other; and a post mount coupled to the wood post bracket, the post mount comprising a base plate secured to the mounting plate of the wood post bracket, and a support pipe extending upward from the base plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The teachings of the present disclosure can be readily understood by considering the following detailed description in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates an isometric view of one example of a bracket according to the present disclosure;

(3) FIGS. 2A-2E are isometric views illustrating the installation of a post using the bracket of FIG. 1;

(4) FIGS. 3A-3E are cross sectional views of certain steps of the installation illustrated in FIGS. 2A-2E, taken along line A-A' of FIG. 2B;

(5) FIG. 4 illustrates several example installation locations of the bracket of FIG. 1;

(6) FIG. 5 illustrates a cross sectional view of an alternate example of the bracket;

(7) FIG. 6 illustrates an isometric view of another example of a bracket according to the present disclosure;

(8) FIGS. 7A-7D are side views illustrating the installation of a post mount using the bracket of FIG. 6; and

(9) FIG. 8 illustrates a top view of the fully installed post mount of FIGS. 7A-7D, using the wood

post bracket of FIG. 6 to reinforce.

(10) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures.

DETAILED DESCRIPTION

(11) The present disclosure describes a novel bracket and a method for securing wood posts in a decking system. In one example, a wood post bracket secures a wood post directly to the rim joist, to the reinforcing block, and to the substructure of the deck's wood framing system. The bracket thus allows the rim joist, reinforcing block, and deck framing to work together to reinforce the vertical wood post quickly and easily with a code compliant installation.

(12) In another example, a wood post bracket can be used to secure the base plate of a post mount to the deck's wood framing system, while also reinforcing the substructure and rim joist of the wood framing system.

(13) As discussed above, one common style of decking system comprises vertically oriented posts that are supported by a wood substructure or framing system that includes joists. The posts are typically secured to the joists by fasteners (e.g., lag bolts or lag screws) which are drilled directly through the posts and joists, without any pre-drilled holes being formed in either the posts or the joists.

(14) Current building codes require that outward movement of a post (i.e., movement in a direction away from the joist to which the post is secured) is not greater than a threshold. Conventionally, the outward movement of the post may be limited by securing a reinforcing block to the post, on the opposite side of the post from the joist. For instance, the fastener may be lagged through the joist, post, and reinforcing block, so that the post is held between the joist and the reinforcing block. However, building codes that are scheduled for implementation in the coming years not only further limit the permitted outward movement of the post, but also limit the permitted inward movement (i.e., movement in a direction toward the joist to which the post is secured) and lateral movement (i.e., sideways movement of the post along the joist to which the post is secured). Although a reinforcing block as described above may provide adequate protection against outward movement of a post, and may also provide some protection against inward movement, the reinforcing block does little to limit the lateral movement of the post. Thus, conventional methods of securing posts to joists are unlikely to ensure compliance with the expected changes in the building codes.

(15) Examples of the present disclosure provide a bracket that may be installed on and under the rim joist of a decking system and a reinforcing block. The bracket includes a sleeve or pocket into which the post may be inserted, and includes apertures so that fasteners may be driven through the rim joist, sleeve, post, and reinforcing block. The bracket acts as a counter pivot to loads exerted on the post in all directions, so that outward, inward, and lateral movement of the post is minimized. The bracket allows the full width and height of the rim joist and the reinforcing block to work together as a counter balance, pushing and pulling in opposite directions to transfer and distribute pressures and loads directly to the deck substructure, as opposed to using just the width of the rim joist for structural reinforcement of the post.

(16) FIG. 1 illustrates an isometric view of one example of a bracket **100** according to the present disclosure. It should be noted that FIG. 1 is not necessarily to scale (e.g., some dimensions may be exaggerated to better show the features of the bracket **100**). In one example, the bracket **100** is formed as a single, unitary piece fabricated from stamped metal. In other words, no fasteners are required to hold the different components or sections of the bracket **100** together.

(17) As illustrated, the bracket **100** generally takes the form of an upside down “T.” Specifically, the bracket **100** comprises a planar base **102** having a generally rectangular shape.

(18) At the center of the base **102** is a hollow sleeve **104** that extends from the base **102** at approximately a ninety degree angle (i.e., perpendicular relative to the base **102**). The sleeve **104** may be sized and shaped to accommodate a wooden post, i.e., such that a wooden post can be

inserted into the sleeve **104**. Thus, in one example, the sleeve **104** may comprise four sides arranged to form a rectangular tube (i.e., a tube having a rectangular cross section) whose dimensions are sized such that a 4 inch by 4 inch post can be held snugly within the hollow interior of the tube.

(19) In one example, two of the four sides of the sleeve **104** (i.e., two non-adjacent sides) each include a gap **112**, such that there are breaks in the perimeter of the sleeve's cross section. In this case, the sleeve **104** may appear to be formed in two halves that are spaced apart from each other, as illustrated in FIG. 1. Forming the gaps **112** in the two sides of the sleeve **104** may reduce the amount of material required to fabricate the bracket **100**. Alternatively or in addition, one or more of the sides of the sleeve **104** may include cutouts or windows cut into the planar surface of the side to further reduce material usage. However, in other examples, the sleeve **104** may be formed to have a continuous perimeter (e.g., without gaps or cutouts in any of the sides). In this case, the sleeve **104** may appear to be formed as a single, solid piece.

(20) A first end **106** of the sleeve **104** is coupled directly to the base **102**, such that the first end **106** of the sleeve **104** is closed. However, a second end **108** of the sleeve **104** defines an opening **110** into which the post may be inserted, as described in further detail below.

(21) Referring back to the base **102**, the base **102** may further include a first end **114** and a second end **116**. The first end **114** and the second end **116** of the base **102** may be spaced equidistant from the sleeve **104** (e.g., such that the sleeve **104** is positioned midway between the first end **114** and the second end **116**). In one example, the first end **114** comprises a first flange **118** that extends from the base **102** at approximately a ninety degree angle (i.e., perpendicular relative to the base **102**, or parallel relative to the sleeve **104**). Similarly, the second end **116** may comprise a second flange **120** that extends from the base **102** at approximately a ninety degree angle (i.e., perpendicular relative to the base **102**, or parallel relative to the sleeve **104** and first flange **118**). The first flange **118** and the second flange **120** may extend from the base **102** in the same direction as the sleeve **104**. In one example, at least one of the first flange **118** and the second flange **120** may be hinged, e.g., such that the first flange **118** and the second flange **120** can rotate or bend to some degree around a respective bending axis **136** and **138**, where the bending axes **136** and **138** run substantially parallel to the first flange **118** and the second flange **120**.

(22) In one example, the bracket **100** may further comprise a plurality of tabs **122** coupled to the perimeter of the second end **108** of the sleeve **104**. In one example, a tab **122** may extend from each corner of the second end **108** of the sleeve **104**, and the tabs **122** may be coupled to non-adjacent sides of the sleeve **104**. For instance, in one example, a first tab **122** and a second tab **122** may extend outward from the sleeve **104**, in a direction toward the first end **114** of the base **102**. Similarly, a third tab **122** and a fourth tab **122** may extend outward from the sleeve **104**, in a direction toward the second end **116** of the base **102**. Each tab **122** may have a planar shape. The tabs **122** may be hinged, e.g., such that the tabs **122** can rotate or bend to some degree around respective bending axes **124** and **126**, where the bending axes **124** and **126** run substantially parallel to the first flange **118** and the second flange **120**.

(23) In one example, the bracket **100** further comprises a plurality of apertures. The plurality of apertures may include a first plurality of apertures **128** that is formed in the sides of the sleeve **104**. More specifically, the first plurality of apertures **128** may be formed in the sides of the sleeve **104** that do not have the tabs **122** coupled thereto (e.g., non-adjacent sides). In one example, some apertures **128** of the first plurality of apertures **128** are formed near the first end **106** of the sleeve, while some apertures **128** of the first plurality of apertures are formed near the second end **108** of the sleeve **104**. In the example illustrated in FIG. 1, each side of the sleeve **104** that includes the first plurality of apertures **128** includes four apertures **128**: two apertures **128** formed near the first end **106** of the sleeve **104** and two apertures **128** formed near the second end **108** of the sleeve **104**.

(24) The plurality of apertures may further include a second plurality of apertures **130** that is formed in the sleeve **104**. More specifically, the second plurality of apertures **130** may be formed in

the sides of the sleeve **104** that have the tabs **122** coupled thereto (e.g., non-adjacent sides). In one example, the second plurality of apertures **130** is formed near the first end **106** of the sleeve **104**. In the example illustrated in FIG. 1, each side of the sleeve **104** that includes the second plurality of apertures **130** includes two apertures **130** formed near the first end **106** of the sleeve **104**. The second plurality of apertures **130** may be situated closer to the base **102** than the lower apertures of the first plurality of apertures **128** that are formed near the first end **106** of the sleeve **104**.

(25) The plurality of apertures may further include a third plurality of apertures **132** that is formed in the flanges **118** and **120**. In the example illustrated in FIG. 1, each of the first flange **118** and the second flange **120** includes two apertures **132**. Each aperture of the third plurality of apertures **132** may be collinear with at least one aperture of the second plurality of apertures **130**, e.g., such that a single fastener may pass through both an aperture of the second plurality of apertures and an aperture of the third plurality of apertures **132**.

(26) The plurality of apertures may further include a fourth plurality of apertures **134** that is formed in the tabs **122**. In the example illustrated in FIG. 1, each tab **122** includes one aperture **134**.

(27) FIGS. 2A-2E are isometric views illustrating the installation of a post using the bracket **100** of FIG. 1. As such, FIGS. 2A-2E also serve as a flow diagram for a method of installing a post using the bracket **100** of FIG. 1. Similar reference numerals are used in FIGS. 2A-2E to refer to elements of the bracket **100** that is illustrated in FIG. 1. FIGS. 3A-3E are cross sectional views of certain steps of the installation illustrated in FIGS. 2A-2E, taken along line A-A' of FIG. 2B.

(28) As illustrated in FIG. 2A, a wood frame **200** comprising a plurality of joists may be provided. It should be noted that FIG. 2A may illustrate a portion of the frame **200**; the full frame **200** may be larger than what is illustrated in FIG. 2A. The frame **200** may be provided as part of the substructure or support system for a deck, where each joist is one of a plurality of horizontal members that is arranged to support the deck. In one example, the plurality of joists may include rim joists **202**, which form part of the perimeter of the frame **200** (e.g., the outside frame of the deck). For instance, FIG. 2A illustrates four rim joists **202** that are spaced apart from each other in a parallel manner. In addition, the plurality of joists may also include a plurality of deck joists **204** positioned inside the perimeter defined by the rim joists.

(29) As illustrated in FIG. 2B, which shows a close-up view of a portion of location **206** of the frame **200** of FIG. 2A, a bracket **100** may be installed on and under one of the rim joists **202**. FIG. 3A illustrates a cross sectional view of this step. The bracket **100** may be configured in a manner similar to the bracket **100** of FIG. 1. In one example, the first end **116** of the base **102** of the bracket **100** may be slid beneath the rim joist **202**, such that the rim joist **202** sits between the sleeve **104** and the first flange **118**. In one example, where a double rim joist is used (e.g., such that two rim joists **202** are positioned flush against each other, the first flange **118** may be folded down (e.g. so that the first flange **118** is substantially coplanar with the base **102**). In this case, the first flange **118** will rest along the bottoms of both rim joists. Two of the tabs **122** at the second end **108** of the sleeve **104** may be folded over the top of the rim joist **202**.

(30) As illustrated in FIG. 2C and FIG. 3B, fasteners may next be lagged through at least some of the apertures in the bracket **100**, in a direction toward (or perpendicular relative to) the base **102**. For instance, fasteners **208** may be lagged through the apertures **134** in the tabs **122** that are folded over the top of the rim joist **202**. This partially secures the bracket **100** to the rim joist **202**.

(31) FIG. 3E illustrates an alternate example of the step shown in FIGS. 2C and 3B, where the rim joist **202** may be larger than shown in FIGS. 2C and 3B. For instance, if the rim joist **202** illustrated in FIGS. 2C and 3B is a 2 inch by 8 inch rim joist, then the rim joist **202** illustrated in FIG. 3E may be larger than an 8 inch rim joist. In the case of FIG. 3E, the tabs **122** at the second end **108** of the sleeve **104** may not be folded over the top of the rim joist **202**, but may instead rest flush against the side of the rim joist **202** as shown. Fasteners **208** may be lagged through the apertures **134** in the tabs **122** in a direction toward the first flange **118** (or parallel relative to the base **102**).

(32) As illustrated in FIG. 2D and FIG. 3C, a reinforcing block **214** may next be installed in the

frame **200**. In one example, the reinforcing block **214** is installed so that the reinforcing block **214** sits between the sleeve **104** and the second flange **120**. Although not shown in FIG. 2D and FIG. 3C, the reinforcing block **214** may be secured with fasteners to the frame **200** (e.g., the ends of the reinforcing block **214** may be secured to two respective deck joists, or a first end of the reinforcing block **214** may be secured to one deck joist while a second end of the reinforcing block **214** is secured to one rim joist, depending upon the location of installation). Two of the tabs **122** at the second end **108** of the sleeve **104** may be folded over the top of the reinforcing block **214**. Fasteners **216** may be lagged through the apertures **134** in the tabs **122** that are folded over the top of the reinforcing block **214**. This partially secures the bracket **100** to the reinforcing block **214**. Moreover, the reinforcing block **214** ties the bracket **100** directly to the frame **200**, as the bracket **100** is now secured on both ends to the rim joist **202** for counterbalance reinforcement and energy dispersal throughout the frame **200**.

(33) As illustrated in FIG. 2E and FIG. 3D, a post (e.g., a 4 inch by 4 inch pressure treated post) may be inserted into the sleeve **104**. Fasteners **222** may secure the post **220** to the rim joist **202**. In this case, the fasteners **222** may be lagged through the apertures **132** in the first flange **118**, through the rim joist **202**, through the apertures **130** in the sleeve **104**, and into the post **220**, as shown. In addition, fasteners **218** may also be lagged through the apertures **132** in the second flange **220**, through the support block **214**, through the apertures **130** in the sleeve **104**, and into the post **220**, as shown. In some examples, one or more bolts **232** may additionally be lagged all the way through the rim joist **202**, through the gap **112** in one side of the sleeve **104**, through the post **220**, through the gap **112** on the opposite side of the sleeve **104**, and through the reinforcing block **214**.

(34) Optionally, fasteners **224** may secure the post **220** to the deck joist **204** (as shown in FIG. 3D), e.g., if the bracket **100** is installed in a corner of the frame **200** where the rim joist **202** meets a deck joist **204**. In this case, the fasteners **224** may be lagged through the deck joist **204**, through the apertures **128** in the sleeve **104**, and into the post **220**. In an example where the bracket **100** is installed mid-span on the rim joist **202** (e.g., not in a corner where the rim joist **202** meets a deck joist **204**), this step may be skipped.

(35) Thus, as shown in FIG. 3D, the bracket **100** allows the full length of the rim joist **202** and the reinforcing block **214** to work together to counter balance the top pull load on the post **220** in any direction. Specifically, the rim joist **202** and the reinforcing block **214** push and pull in opposite directions to transfer and distribute pressures and loads directly to the deck sub structure, as opposed to using just the width of the rim joist **202** for structural reinforcement of the post. For instance, when a pull force is applied to the post **220** as shown in FIG. 3D, the bracket **100** pulls the rim joist and reinforcing block in opposite directions both up and down (as shown by force arrows F.sub.1 and F.sub.2), as well as in and out.

(36) As discussed above, FIGS. 2A-2D illustrate only one example installation of the bracket **100**. Specifically, FIGS. 2A-2D illustrate an example in which the bracket **100** is installed in a corner of the frame **200** (e.g., where two joists meet). However, the bracket **100** may also be installed to secure a post mid-span on a joist, i.e., in the middle of the joist as opposed to where the joist meets another joist.

(37) FIG. 4 illustrates several example installation locations of the bracket **100** of FIG. 1. For instance, the location **206** illustrates the installation shown in FIG. 2A, where the bracket **100** is installed in a corner of the frame **200** (i.e., where two rim joists **202** meet). In this case, the reinforcing block **214** is installed parallel to one rim joist **202** and perpendicular to the other rim joist **202** (e.g., so that the reinforcing block is secured between one rim joist **202** and an adjacent deck joist **204**).

(38) The location **228**, however, illustrates an example in which the bracket **100** is installed mid-span on a rim joist **202**, between two deck joists **204**. In this case, the reinforcing block **214** is installed parallel to the rim joist **202** (or perpendicular to the deck joists **204** to which the bracket **100** is secured).

(39) FIG. 5 illustrates a cross sectional view of an alternate example of the bracket **100**. Similar reference numerals to those used in the previous figures are used for similar components in FIG. 5. In the example illustrated in FIG. 5, the planar base **102** is shortened, such that the base **102** extends to a single side of the sleeve **104**.

(40) More particularly, the first flange **118** and the portion of the base **102** connecting the first flange **118** to the sleeve **104** have been removed in FIG. 5. In this case, the bracket **100** is secured to the rim joist **202** by fasteners **208** that are driven through the tabs **122** (e.g., through apertures of the fourth plurality of apertures **134**). The bracket **100** may be further secured to the rim joist **202** by fasteners **234** that are driven through apertures (e.g., of the second plurality of apertures **130**) in the side of the sleeve **104** that is positioned flush against the rim joist **202**. Thus, the bracket **100** of FIG. 5 does not wrap around the bottom of the rim joist **202**.

(41) The bracket **100** is secured to the reinforcing block **214** as described above.

(42) The location **230** illustrates another example in which the bracket is installed mid-span on a rim joist **202**. However, in this case, the bracket **100** is situated between two joists **226** that are secured between the rim joist **202** and an adjacent deck joist **204**. In this case, the reinforcing block **214** is installed parallel to the rim joist **202** (or perpendicular to the joists **226** to which the bracket **100** is secured).

(43) In another example, a wood post bracket can be used to secure the base plate of a post mount to the deck's wood framing system, while also reinforcing the substructure and rim joist of the wood framing system. In some examples, rather than placing a wood post within the hollow sleeve of the wood post bracket and securing the wood post directly to the substructure of the deck's wood framing system, a post mount may instead be secured to the top of the deck's wood framing system. The post mount may comprise a support pipe (e.g., a metal post or tube) over which a decorative sleeve may be placed, and may include a planar base plate for attachment to the top of the deck's wood framing system. Conventionally, a plurality of (e.g., three) wood mounting blocks help couple the post mount to the rim joist. However, according to one example of the present disclosure, a wood post bracket such as that described above (or slightly modified) may be used to reinforce the substructure and rim joist of the wood framing system while also providing a means of securing the post mount to the wood framing system.

(44) FIG. 6 illustrates an isometric view of another example of a bracket **600** according to the present disclosure. It should be noted that FIG. 6 is not necessarily to scale (e.g., some dimensions may be exaggerated to better show the features of the bracket **600**). In one example, the bracket **600** is formed as a single, unitary piece fabricated from stamped or extruded metal (e.g., aluminum). In other words, no fasteners are required to hold the different components or sections of the bracket **600** together.

(45) As illustrated, the bracket **600** generally takes the form of “J.” Specifically, the bracket **600** comprises a planar base **602** having a generally rectangular shape.

(46) At one end of the base **602** is a hollow sleeve **604** that extends from the base **602** at approximately a ninety degree angle (i.e., perpendicular relative to the base **602**). The sleeve **604** may be sized and shaped to accommodate a wood mounting block for use in securing a post mount (e.g., a 2”×6” board), i.e., such that a wood mounting block can be inserted through the sleeve **604** (e.g., in the direction of the arrow **630**). Thus, in one example, the sleeve **604** may comprise four walls arranged to form an open rectangular passage (i.e., an open passage having a rectangular cross section) whose dimensions are sized such that a 2”×6” board can be held snugly within the hollow interior of the passage.

(47) A first end **606** of the sleeve **604** is coupled directly to the base **602**, while a second end **608** of the sleeve **604** is coupled to a mounting plate **610**. Thus, the sleeve **604** is positioned between the base **602** and the mounting plate **610**. Moreover, the base **602** and the mounting plate **610** are oriented substantially parallel to each other. Thus, one wall of the open rectangular passage formed by the sleeve **404** may be bounded by the base **602**, while an opposite wall of the open rectangular

passage may be bounded by the mounting plate **610**.

(48) Referring back to the base **602**, the base **602** may include a first flange **614** that extends away from the base **602** in an orientation that is parallel to the sleeve **604**, such that the base **602** as a whole has an “L” shaped cross section. The first flange **614** may include a first plurality of (e.g., two) apertures **616** formed therein.

(49) The mounting plate **610** may have a generally rectangular shape and may include a second flange **618** that extends away from the mounting plate **610** and toward the first flange **614** in an orientation that is parallel to the sleeve **604**, such that the mounting plate **610** as a whole, like the base **602**, has an “L” shaped cross section. Thus, collectively, the base **602** (i.e., including the first flange **214**) and the mounting plate **610** (i.e., including the second flange **618**) may form a rectangular channel **624** with an outer surface of the sleeve **604**. The rectangular channel **624** may extend in a parallel manner relative to the sleeve **604** and may be of similar dimensions to the sleeve **604** (e.g., equal width, length, and depth). Thus, the rectangular channel **624** may be sized such that a 2”×6” board can be held snugly within the hollow interior of the rectangular channel **624**.

(50) Referring back to the mounting plate **610**, the second flange **618** may include a second plurality of (e.g., two) apertures **620** formed therein. Additionally, a third plurality of (e.g., four) apertures **622** may be formed in the surface of the mounting plate **610** that is parallel to the base **602**. The spacing between the apertures **622** of the third plurality of apertures **622** may be arranged to mirror the spacing of apertures in the base plate of a post mount, as described in further detail below. Thus, in one example, the apertures **622** of the third plurality of apertures **622** may be arranged such that each aperture **622** is positioned at a different corner of a common rectangle or square. However, other spacings and arrangements of the apertures **622** are also possible based on the configuration of the post mount's base plate.

(51) The mounting plate **610** may be longer than the base **602**. That is, while the end of the base **602** opposite the flange **612** may end flush with the sleeve **604**, the end of the mounting plate **610** opposite the flange **618** may extend past the sleeve **604** to form a lip **626**.

(52) FIGS. 7A-7D are side views illustrating the installation of a post mount using the bracket **600** of FIG. 6. As such, FIGS. 7A-7D also serve as a flow diagram for a method of installing a post mount using the bracket **600** of FIG. 6. Similar reference numerals are used in FIGS. 7A-7D to refer to elements of the bracket **600** that is illustrated in FIG. 6.

(53) As illustrated in FIG. 7A, a first wood mounting block **700**, such as a 2”×6” board, may be inserted into the rectangular channel **624** of a wood post bracket **600** (e.g., following the direction of the arrow **630** of FIG. 6). As discussed above, the rectangular channel **624** may be defined by the base **602**, the mounting plate **610**, and the outer surface of the sleeve **604** of the wood post bracket **600**. The first wood mounting block **700** may be secured to the wood post bracket **600** by inserting a first plurality of fasteners **702** (e.g., lag screws) through the first plurality of apertures **616** of the wood post bracket **600** and into the first wood mounting block **700** and inserting a second plurality of fasteners (e.g., lag screws) **704** through the second plurality of apertures **720** of the wood post bracket **600** and into the first wood mounting block **700**. For instance, a lag screw may be inserted through each aperture of the first plurality of apertures **616** and the second plurality of apertures **620** and lagged into the wood of the first wood mounting block **700**.

(54) Optionally, a second wood mounting block **706** may also be inserted into the sleeve **604** of the wood post bracket **600** (e.g., following the direction of the arrow **630** of FIG. 6). However, in some examples, the sleeve **604** may be left empty (i.e., no wood mounting block may be inserted in the sleeve **604**). The second wood mounting block **706** may have the same dimensions as the first wood mounting block **700** (e.g., may comprise a 2”×6” board).

(55) As illustrated in FIG. 7B, the wood post bracket **600** may next be positioned such that the lip **626** of the mounting plate **610** rests along the top of a third wood mounting block **708**. The third wood mounting **708** block may, in turn, be secured to a rim joist **710** of a wood deck framing

system (e.g., similar to the wood frame **200** of FIGS. 2A-2E).

(56) As illustrated in 7C, a post mount **712** may next be placed over the mounting plate **610** of the wood post bracket **600**. As discussed above, the post mount **712** may include a base plate **714** and a support pipe **716** extending upward from the base plate **714**. The base plate **714** may include a plurality of (e.g., four) apertures (not shown). The base plate **714** may be positioned such that each aperture in the base plate **714** is aligned with an aperture **622** of the third plurality of apertures **622** in the mounting plate **610**.

(57) The post mount **712** may be secured to the wood post bracket **600** (and, thus, secured to the wood deck framing system) by inserting a third plurality of fasteners (e.g., lag screws) **718** through the apertures in the base plate **714**, through the third plurality of apertures **622** of the mounting plate **610**, and into the first or third wood mounting block **700** or **708** (depending upon the locations of the apertures). For instance, a lag screw may be inserted through each aperture of the base plate **714** and the third plurality of apertures **622** and lagged into the wood of the first or third wood mounting block **700** or **708**. In one example, a decking surface material (i.e., the visible surface material of the deck structure, as opposed to the invisible, supporting substructure of framing) may be positioned between the mounting plate **610** of the wood post bracket **600** and the base plate **714** of the post mount **712**.

(58) FIG. 7D illustrates the post mount **712** fully installed using the wood post bracket **600** to reinforce. FIG. 8 illustrates a top view of the fully installed post mount **712** of FIGS. 7A-7D, using the wood post bracket **600** of FIG. 6 to reinforce. In other words, FIG. 8 represents a top view of FIG. 7D. FIG. 8 additionally shows how the rim joist **710**, first wood mounting block **700**, second wood mounting block **706**, and third wood mounting block **708** are arranged relative to other portions of a deck's wood framing system, namely, deck joists **720**.

(59) Thus, the wood post bracket **600** allows the full length of the rim joist **710** and the wood mounting blocks **700**, **706**, and **708** to work together to counter balance the top pull load on the post mount **712** in any direction. Specifically, the rim joist **710** and the wood mounting blocks **700**, **706**, and **708** push and pull in opposite directions to transfer and distribute pressures and loads directly to the deck sub structure, as opposed to using just the width of the rim joist **710** for structural reinforcement of the post mount **712**. Thus, the wood post bracket **400** provides a stable installation platform for the post mount **712** while also reinforcing the deck substructure.

(60) Although various embodiments which incorporate the teachings of the present disclosure have been shown and described in detail herein, those skilled in the art can readily devise many other varied embodiments that still incorporate these teachings. In addition, while various embodiments have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of a claimed embodiment should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. An apparatus, comprising: a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls; a planar base forming a first wall of the four walls, wherein the planar base has a rectangular shape that includes a first flange that extends away from the planar base in an orientation parallel to the hollow sleeve, such that the planar base has an L-shaped cross section; and a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate and the planar base and the mounting plate are oriented parallel to each other, wherein the mounting plate has a rectangular shape that includes a second flange that extends away from the mounting plate in a direction toward the first flange and in an orientation parallel to the hollow sleeve, such that the mounting plate has an L-shaped cross section.

2. The apparatus of claim 1, wherein the open rectangular passage is sized to accommodate a two inch by six inch board.
3. The apparatus of claim 1, wherein the first flange includes a plurality of apertures.
4. The apparatus of claim 1, wherein a rectangular channel is defined by the first flange, the second flange, and an outer surface of the hollow sleeve.
5. The apparatus of claim 4, wherein the rectangular channel is sized to accommodate a two inch by six inch board.
6. The apparatus of claim 1, wherein the second flange includes a plurality of apertures.
7. The apparatus of claim 1, wherein the mounting plate includes a lip that extends past the hollow sleeve, such that the mounting plate is longer than the planar base.
8. The apparatus of claim 7, wherein the lip is positioned on an opposite end of the mounting plate from the second flange.
9. The apparatus of claim 1, wherein the mounting plate includes a plurality of apertures formed in a surface of the mounting plate that is oriented parallel to the planar base.
10. The apparatus of claim 9, wherein the plurality of apertures are arranged such that each aperture of the plurality of apertures is positioned at a different corner of a common rectangle.
11. The apparatus of claim 1, wherein the apparatus is formed from a single piece of extruded aluminum.
12. An apparatus, comprising: a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls; a planar base forming a first wall of the four walls, wherein the planar base comprises: a first flange that extends away from the planar base in an orientation parallel to the hollow sleeve, such that the planar base has an L-shaped cross section; and a first plurality of apertures formed in the first flange; and a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate, wherein the mounting plate comprises: a second flange that extends away from the mounting plate in a direction toward the first flange and in an orientation parallel to the hollow sleeve, such that the mounting plate has an L-shaped cross section; a second plurality of apertures formed in the second flange; a third plurality of apertures formed in a surface of the mounting plate that is oriented parallel to the planar base; and a lip positioned on an opposite end of the mounting plate from the second flange and extending past the hollow sleeve, such that the mounting plate is longer than the planar base.
13. The apparatus of claim 12, wherein the hollow sleeve, the planar base, and the mounting plate form a J shape.
14. The apparatus of claim 12, wherein the apparatus is formed from a single piece of extruded aluminum.
15. The apparatus of claim 12, wherein a rectangular channel is defined by the first flange, the second flange, and an outer surface of the hollow sleeve.
16. The apparatus of claim 15, wherein each of the open rectangular passage and the rectangular channel is sized to accommodate a two inch by six inch board.
17. The apparatus of claim 12, wherein the second plurality of apertures arranged such that each aperture of the second plurality of apertures is positioned at a different corner of a common rectangle.
18. The apparatus of claim 12, wherein the lip is positioned on an opposite end of the mounting plate from the second flange.
19. A deck substructure, comprising a rim joist coupled between a first deck joist and a second deck joist; a wood post bracket coupled to the rim joist, the wood post bracket comprising: a hollow sleeve defining an open rectangular passage, the open rectangular passage having four walls; a planar base forming a first wall of the four walls; a mounting plate forming a second wall of the four walls that is positioned opposite the first wall, such that the hollow sleeve is positioned between the planar base and the mounting plate and the planar base and the mounting plate are

oriented parallel to each other; and a post mount coupled to the wood post bracket, the post mount comprising: a base plate secured to the mounting plate of the wood post bracket; and a support pipe extending upward from the base plate.

20. The deck substructure of claim 19, wherein the open rectangular passage is sized to accommodate a two inch by six inch board.
